Reading Between The Lines: Modeling and Evaluating Behavioral Realism in Legal Simulation

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Code: `github.com/liftlab-SLS/reading-between-the-lines`

Abstract

Deposition training requires attorneys to recognize and manage dynamic interpersonal behaviors that existing legal-AI benchmarks do not measure. We introduce Witness Sim, a generative deposition simulator, and evaluate its behavioral realism using emotion-trajectory analysis. By projecting language-model activations onto 23 directional emotion vectors, we compare temporal affective dynamics between 55 real deposition transcripts and synthetic depositions generated across ten witness archetypes. Synthetic depositions reproduce statistically significant temporal emotional structure found in real testimony but exhibit systematically compressed emotional intensity and remain distinguishable from real depositions in trajectory-level analyses. Further, we find that persistent witness dispositions and transient behavioral shifts operate on distinct temporal scales. These results suggest that emotion trajectories provide a principled metric for evaluating behavioral realism in legal simulation and highlight temporal behavioral dynamics as an important, underexplored evaluation axis for LLM-based simulations.

1. Introduction

Depositions are a strategically critical phase of litigation, yet training attorneys to conduct them is difficult to scale and evaluate. Junior attorneys typically learn by observing senior litigators, through occasional mock exercises, or through conducting real depositions. However, real depositions leave little room for trial-and-error, observation offers no active practice, and structured mock sessions require substantial time from experienced attorneys or live witnesses. These constraints can be especially acute outside resource-rich environments. Effective examiners must also do more than extract facts from a witness: they probe consistency, surface omissions, manage resistance, and adapt as a witness becomes evasive, defensive, or fatigued. These dynamic interpersonal behaviors are central to deposition practice but largely absent from current benchmarks for legal AI. We introduce Witness Sim, a generative deposition-training system and testbed for studying these dynamics. Witness Sim instantiates each witness from underlying case materials and assigns a behavioral archetype, such as cooperative, defensive, evasive, or combative. As questioning progresses, the platform updates a six-dimensional psychological state through coupled ODEs driven by examiner pressure and topic sensitivity. This dynamical structure is designed for controllability rather than relying on an opaque learned recurrent state, with the simulator exposing low-dimensional variables that can be adjusted.

The central question, however, is evaluative: what constitutes behavioral realism for simulated witnesses in legal training environments? Existing legal-AI evaluations tend to emphasize factuality and reasoning. These measures are necessary but incomplete for deposition training, where realism depends on how witness behavior changes over time in response to questioning. Thus, we treat deposition simulation as a legal evaluation tool- a useful simulator must preserve case facts while also reproducing the behavioral dynamics present during testimony.

To evaluate these dynamics, we adapt emotion-vector analysis from recent interpretability work on affective and interpersonal concepts in language-model activations (Sofroniew et al., 2026). We do not treat these vectors as evidence that a model experiences emotion; instead, we use them as measurable proxies for affective and interactional structure in text. Our evaluation passes both real and synthetic deposition transcript segments through an analysis model and projects hidden activations onto emotion-concept vectors. Because this measurement is separate from Witness Sim's internal ODE state, it tests an external behavioral trace rather than directly reading the simulator's control variables.

Our empirical study draws on a corpus of 55 real deposition transcripts from the National Prescription Opiate Litigation (Case No. 1:17-MD-2804), a federal multidistrict litigation consolidating cases against opioid manufacturers, distributors, and pharmacies, and a matched set of synthetic depositions generated under different witness personas and questioning strategies using Claude-haiku-4.5. We investigate the following research axes: (1) do configurable witness personas produce distinguishable affective trajectories during simulated depositions?; (2) do different questioning strategies induce predictable transitions in witness affective and interpersonal state?; and (3) can emotion-vector trajectories provide a quantitative realism metric that distinguishes real and synthetic depositions and identifies calibration gaps in LLM-based legal simulations?

Accordingly, we make the following contributions. We formulate deposition realism as a dynamic legal-evaluation problem rather than a static question-answering problem. We introduce an emotion-vector evaluation framework for measuring the behavioral realism of LLM-based legal simulations, using deposition practice as a testbed. We present an empirical comparison using Witness Sim to show that simulated witnesses can approximate the direction of real deposition affective dynamics. More broadly, we argue that emotional and interpersonal presentation is an underexplored axis of evaluation for LLM-based simulations: high fidelity legal simulation depends not only on producing legally and factually correct text, but also on being able to model human behavior and emotional response.

2. Related Work

This work connects three lines of research that have largely progressed independently: affective representations in language models, conversation-level emotion modeling, and LLM-based simulation for professional training. Existing work provides tools for quantifying affect, characterizing how it evolves with interactional context, and generating plausible simulated encounters. However, these capabilities have not been integrated to assess whether simulated professional interactions reproduce the behavioral dynamics observed in real-world settings. We address this gap in the domain of deposition training, where emotional and interpersonal dynamics are not ancillary details, but core determinants of both simulation fidelity and pedagogical value.

Affective computing and emotion in language models. Work in affective computing established that emotion can be systematically recognized, modeled, and simulated by computational systems rather than treated as an informal feature of human interaction. Picard (1997) argues that computers intended to interact naturally with humans must be able to interpret and respond to affective states. Recent language-model interpretability work provides a more specific technical basis: Sofroniew et al. (2026) shows that language models contain structured internal representations of emotion concepts—"emotion vectors"—that activate in emotionally meaningful contexts and can influence model behavior. If emotional and interpersonal states are represented in model activation space, then emotion-vector analysis can be used to evaluate whether a simulated deposition produces affective dynamics resembling those in real depositions.

Emotion recognition in conversation. Rather than treating emotion as isolated incidents, recent emotion recognition in conversation (ERC) models show that emotional state must be inferred from speaker role, turn history, and interpersonal context. DialogueGCN models conversation-level emotion through both self-speaker and inter-speaker dependencies, showing that affect depends on how utterances build on prior turns (Ghosal et al., 2019). This is directly relevant to depositions, where defensiveness, composure, or evasiveness often emerges from accumulated attorney pressure rather than a single answer. EmoBERTa motivates explicitly marking speaker roles, since the same language can signal different affective states depending on whether it is produced by the attorney or the witness (Kim & Vossen, 2021). DialogXL motivates maintaining longer-range conversational memory, since affective state in deposition can change cumulatively as pressure, contradiction, and recall demands build over time (Shen et al., 2021). Witness Sim adapts these insights to a generative legal-training task, using affective state to condition how a simulated witness responds under adversarial questioning.

LLM-based simulation for professional training.

Recent work shows that language models can generate plausible social behavior when embedded in agent architectures. Generative Agents demonstrates that LLM agents equipped with memory, planning, and reflection can produce believable social interactions in an open-ended simulated environment (Park et al., 2023). This line of work establishes that LLMs can support rich social simulation, but its goal is not to reproduce the behavior of a specific real person or to evaluate whether an agent's behavior follows empirically observed trajectories. The simulation target is believability within an imagined social world, rather than fidelity to a real interactional record.

A second line of work applies LLM-based simulation to professional training. LLM role-play systems have been used to support practice in negotiation, conflict resolution, counseling, and medical education (Yang et al., 2024; Shaikh et al., 2024; Louie et al., 2026; Li et al., 2024). These systems show that simulated interlocutors can scaffold repeated practice and improve trainee performance in interpersonal domains. However, evaluation in these settings typically focuses on learning outcomes, user perceptions, expert ratings, or conversational plausibility. Even when the simulated partner represents a patient, client, or counseling scenario, the central question is usually whether the interaction is useful or realistic enough for training – not whether the simulator reproduces the emotional responses and behavioral dynamics of real human interactions.

Work on persona simulation moves closer to the problem of imitation. Eval4Sim argues that persona-grounded simulations should be evaluated against human conversational behavior rather than relying only on opaque plausibility scores or LLM-as-judge ratings (Bao et al., 2026). This is an important step toward empirical validation of simulated agents. Still, most persona-simulation evaluations remain centered on response-level fidelity: whether a model gives answers consistent with a person's facts, preferences, opinions, or likely conversational responses. Such evaluations capture whether a simulator says the kinds of things a person might say, but they do not directly test whether the simulator behaves over time like a person does in a sustained, adversarial interaction.

Legal simulation work similarly demonstrates the promise of LLMs for modeling legal roles and training environments, but has not yet addressed trajectory-level behavioral realism. AgentCourt, SimCourt, and AI-Assisted Moot Courts simulate courtroom participants, legal argument, or attorney training settings, but their evaluations emphasize legal reasoning, procedural structure, argument quality, and role fidelity rather than interpersonal or emotional dynamics (Chen et al., 2025; Zhang et al., 2025; 2026). LegalBench likewise evaluates legal reasoning capabilities, but does not measure the social, emotional, or behavioral dynamics that shape legal practice (Guha et al., 2023). These systems are valuable for assessing whether models can reason legally or occupy legal roles, but they do not ask whether a simulated witness becomes defensive, evasive, fatigued, combative, or composed in ways that resemble real testimony.

Taken together, this literature shows that LLMs can simulate plausible social agents, support professional role-play, and occupy legal roles in courtroom or training environments. However, these strands of work leave open a distinct evaluation problem. Plausibility is not the same as fidelity to a real interaction, and response-level imitation is not the same as reproducing how behavior unfolds over time. In legal settings, this distinction matters because effective litigation requires more than asking questions or eliciting facts: lawyers must also work with and around a witness's unique demeanor, personality, resistance, and emotional responses during questioning.

Witness Sim addresses this gap by treating deposition simulation as a problem of temporal behavioral realism. Rather than evaluating only whether a simulated witness gives legally or factually appropriate answers, we ask whether its emotional responses and affective trajectory resembles the dynamics observed in real adversarial questioning. This connects affective computing and conversation-level emotion modeling to legal simulation, and provides a quantitative framework for evaluating whether LLM-based witnesses behave realistically across an interaction, not merely whether they produce plausible individual responses.

3. Methods

3.1. WitnessSim

WitnessSim is a deposition simulator built around a continuous dynamical system that models how a witness's psychological state evolves under adversarial questioning. An LLM generates the surface dialogue, but the content of each answer is conditioned on a low-dimensional state vector that is updated deterministically from features of the attorney's question. This decoupling lets us examine, separately from the language model, whether the behavioral trajectories produced by the simulator correspond to measurable emotional structure in real depositions.

3.1.1. State Space and Archetypes

A witness at turn t is represented by a state vector $\mathbf{s}_t \in [0,1]^6$ with components $\mathbf{s} = (C, K, A, V, R, P)$ denoting *composure*, *knowledge*, *agreeableness*, *verbosity*, *rigidity*, and *performance*, respectively. These dimensions function as a loose analog of the Big Five, with composure mapping to inverse neuroticism, agreeableness to its namesake, and rigidity to inverse openness, while knowledge, verbosity, and performance capture deposition-specific behaviors not well covered by the Big Five.

Each of the ten archetypes used in this study (combative, cooperative, defensive, dogmatic, inventive, loquacious, nervous, neutral, overconfident, overprepared) is defined by an attractor state $\mathbf{s}_0 \in [0,1]^6$ toward which the dynamics relax in the absence of pressure. The simulator is initialized with $\mathbf{s}_0$ extracted from case materials by an LLM persona-generation step, and the attractor remains fixed throughout the deposition.

3.1.2. Question Encoding

Before each state update, the attorney's question q_t is encoded into two scalars. The pressure score $p_t \in [0,1]$ is a count of linguistic markers:

$$p_t = \mathrm{clip}_{[0,1]}\Big(0.1 + 0.2 \cdot n_{\text{high}}(q_t) + 0.1 \cdot n_{\text{med}}(q_t)\Big), \quad (1)$$

where n_{high} counts high-pressure phrases (e.g., *"isn't it true," "you knew," "admit"*) and n_{med} counts medium-pressure phrases (e.g., *"why did you," "who authorized"*). These markers were drawn from forensic-interview tactics documented in the interrogation literature.

The sensitivity score $s_t \in [0, 1]$ is computed via Jaccard similarity between the token set of q_t and that of each of the witness's pre-registered sensitive topics $\mathcal{T}$:

$$s_t = \max_{\tau \in \mathcal{T}} \frac{|\mathrm{tok}(q_t) \cap \mathrm{tok}(\tau)|}{}$$

—tok(q$_t$) $\cup\, \mathrm{tok}(\tau)|$, (2) thresholded at 0.15 to suppress incidental overlap. A composite stress signal is defined as $\xi_t = 0.5p_t + 0.3s_t + 0.2\mathbb{1}[q_t \text{ leading}]$.

3.1.3. State Dynamics

The state evolves through six coupled discrete-time updates derived as forward-Euler steps of a first-order ODE system. Each component combines a pressure-driven term with a homeostatic relaxation toward the attractor $\mathbf{s}_0$:

$$\begin{aligned}
\Delta C &= -0.25\,\xi_t C_t + 0.08(C_0 - C_t),\\
\Delta K &= -0.30\,s_t(K_t - 0.1)\\
&\quad + 0.06(1 - p_t)(K_0 - K_t),\\
\Delta A &= -0.15\,p_t\,\mathrm{sgn}\big(A_t - \tfrac{1}{2}\big)\,\big|A_t - \tfrac{1}{2}\big|\\
&\quad + 0.05(A_0 - A_t),\\
\Delta V &= 0.20\,s_t(1 - V_t) - 0.20\,\mathbb{1}_{\text{lead}} V_t\\
&\quad + 0.05(V_0 - V_t),\\
\Delta R &= 0.12\,p_t(1 - R_t) - 0.03(R_t - R_0),\\
\Delta P &= -0.18\,p_t P_t \phi_t + 0.04(P_0 - P_t).
\end{aligned} \quad (3)$$

where $\phi_t = \min(1, t/20)$ is a fatigue multiplier that imposes a progressive burden on prepared witnesses. After each update, every component is clamped to $[0, 1]$.

Each equation encodes a specific behavioral assumption about how the corresponding state dimension responds to pressure and sensitivity; full justifications are given in Appendix A.

3.1.4. Memory Model

Separately from $\mathbf{s}_t$, each sensitive topic τ carries a recall-quality scalar $\rho_\tau \in [0.1, 1.0]$:

$$\rho_\tau \leftarrow \begin{cases} \max\big(0.1,\ \rho_\tau - 0.25\,p_t\big) & \text{if } \tau \text{ is hit by } q_t,\\ \min\big(1.0,\ \rho_\tau + 0.02\big) & \text{otherwise.} \end{cases} \quad (4)$$

Event	Condition
Witness rattled	$p_t > 0.58 \wedge (C_t < 0.38 \vee \Delta C_t < -0.05)$
Witness combative	$A_t < 0.28 \wedge R_t > 0.60$
Attorney interrupts	$p_t > 0.45 \wedge \lvert q_t \rvert \le 8$ words $\wedge$ $\mathrm{type}(q_t) \in \{\text{closed, leading}\}$
Witness talks over	$R_t > 0.72 \wedge V_t > 0.58 \wedge p_t > 0.35$
Personality shift	$\lVert \mathbf{s}_t - \mathbf{s}_0 \rVert_\infty > 0.35$

Table 1. Threshold conditions on the state vector for discrete behavioral events. Triggered events are passed to the LLM system prompt to ground generated dialogue in the current mathematical state.

Each topic also has an intrinsic sensitivity weight σ_τ set at persona generation. This local memory channel produces realistic inconsistency accumulation: witnesses pressed repeatedly on the same topic become progressively less reliable on it, independent of their global composure.

3.1.5. Derived Behavioral Scores

Four interpretable composites are computed each turn from $\mathbf{s}_t$:

$$\begin{aligned}
\text{consistency}_t &= 0.5K + 0.4R - 0.1(1 - K)(1 - R),\\
\text{evasion}_t &= 0.5(1 - K) + 0.3|2V - 1| + 0.2P,\\
\text{realism}_t &= \max\big(0,\ 1 - 0.3|2C - 1| - 0.3|2K - 1|\\
&\qquad - 2.0\max(0, P - 0.8)\big),\\
\text{adversarial}_t &= 0.4(1 - K) + 0.3(1 - A)\\
&\qquad + 0.2R + 0.1(1 - C).
\end{aligned} \quad (5)$$

The interaction term in consistency penalizes the joint failure mode of low knowledge and low rigidity (an inconsistent and pliable witness). Realism is penalized by extremes in C and K — a witness who is either perfectly calm or perfectly rattled reads as unrealistic — and heavily penalized once performance exceeds 0.8, the regime where preparation reads as coaching. Adversarial weights poor recall and hostility equally, treating a forgetful witness as structurally adversarial regardless of stated agreeableness.

3.1.6. Event Detection

A small set of discrete behavioral events is fired when threshold conditions on the state vector are met (Table 1). The triggered events, along with the current state, are appended to the LLM system prompt at generation time, anchoring the surface dialogue in the underlying dynamics rather than allowing the LLM to drift.

3.1.7. GENERATION LOOP

For each deposition turn, WitnessSim (i) encodes the attorney question into (p_t, s_t, ξ_t), (ii) updates the state vector and per-topic recall, (iii) computes the four derived scores and evaluates the event predicates, and (iv) prompts the underlying LLM with the persona, current state, fired events, and conversation history to generate the witness's answer. The state vector and event flags are logged at every turn, yielding the trajectory data analyzed in our experimental section.

3.2. Emotion Vectors

3.2.1. EMOTION VECTOR CONSTRUCTION

We constructed 23 directional emotion vectors in the residual stream of Llama-3.1-8B-Instruct following the methodology of Sofroniew et. al. For each of 23 target emotions — including *anxious*, *hostile*, *calm*, *resigned*, *suspicious*, *warm*, *confident*, and *defiant*, among others — we generated 125 short narrative stories (2–4 paragraphs each) using Claude-opus-4.5 via the Anthropic API. Stories were prompted to evoke the target emotional state without naming it directly (see Appendix), and were generated across 25 topically diverse scenarios to prevent topic-emotion confounding, yielding approximately 2,875 emotion stories total.

Each story was passed through Llama-3.1-8B-Instruct and hidden states were extracted from layer 21 (of 32 total), again following prior work done by Sofroniew et. al, who identify mid-to-late residual stream layers as the locus of structured affective representations in transformer language models. States were mean-pooled over token positions 50 and onward to avoid prompt-format artifacts. The raw emotion vector for each emotion e was computed as the mean activation across all 125 stories, with the cross-emotion mean subtracted for centering:

$$\boldsymbol{\mu}_e = \frac{1}{N}\sum_{i=1}^{N}\mathbf{h}_i^{(e)}, \qquad \tilde{\mathbf{v}}_e = \boldsymbol{\mu}_e - \frac{1}{23}\sum_{e'=1}^{23}\boldsymbol{\mu}_{e'}, \quad (6)$$

where $N = 125$ and $\mathbf{h}_i^{(e)}$ is the mean-pooled layer-21 hidden state for story i of emotion e.

3.2.2. DENOISING

To remove variance shared across all emotional states we generated 250 neutral stories and applied PCA to their layer-21 activations. The top K principal components explaining 50% of neutral-story variance were projected out of each emotion vector:

$$\mathbf{v}_e = \left(\mathbf{I} - \mathbf{U}_K\mathbf{U}_K^\top\right)\tilde{\mathbf{v}}_e, \qquad \hat{\mathbf{v}}_e = \frac{\mathbf{v}_e}{\|\mathbf{v}_e\|_2}, \quad (7)$$

where $\mathbf{U}_K \in \mathbb{R}^{4096\times K}$ contains the top K principal components of neutral-story activations. Validation on held-out stories (40 per emotion) confirmed that denoising improved discrimination: mean rank of the correct emotion improved from chance (12/23) to 3.9/23.

3.2.3. EMOTION ARC EXTRACTION

Each deposition turn was encoded through Llama-3.1-8B-Instruct using the same procedure: layer-21 hidden states were mean-pooled from token position 50 onward, producing a single 4096-dimensional vector per turn:

$$\mathbf{h}_t = \frac{1}{|\mathcal{T}_t|}\sum_{\tau\in\mathcal{T}_t}\mathbf{H}_\tau^{(21)}, \qquad \hat{\mathbf{h}}_t = \frac{\mathbf{h}_t}{\|\mathbf{h}_t\|_2}, \quad (8)$$

where $\mathcal{T}_t$ denotes token positions from position 50 onward for turn t.

This vector was cosine-projected onto each of the 23 denoised emotion vectors, yielding a 23-dimensional emotion score vector per turn:

$$s_{t,e} = \hat{\mathbf{h}}_t\cdot\hat{\mathbf{v}}_e \in [-1, 1], \qquad \mathbf{s}_t = (s_{t,1}, \ldots, s_{t,23}) \in \mathbb{R}^{23}. \quad (9)$$

Attorney and witness turns were encoded and analyzed separately.

To compare emotion trajectories across depositions of varying length, each deposition was normalized to a 0–100% positional scale and interpolated to $B = 20$ uniform bins:

$$\tilde{s}_{b,e} = \mathrm{interp}\left(\frac{b-1}{B-1};\ \frac{t-1}{T-1},\quad s_{t,e}\right), \quad b \in \{1,\ldots,B\}. \quad (10)$$

The resulting arc matrix $\tilde{\mathbf{S}} \in \mathbb{R}^{B\times 23}$ is flattened to a 460-dimensional arc vector $\mathbf{a} = \mathrm{vec}(\tilde{\mathbf{S}}) \in \mathbb{R}^{460}$, which serves as the base representation for all subsequent correlation and PCA analyses.

3.3. Experimental Setup

3.3.1. FEATURE REPRESENTATIONS

All experiments operate on two parallel turn-level feature sets derived from the 100 synthetic transcripts (10 witnesses $\times$ 10 archetypes, 67,956 turns total).

Emotion features. For each turn t, the 23-dimensional cosine-projected emotion score vector $\mathbf{s}_t \in \mathbb{R}^{23}$ described in Section 3.2.

Residual features. For each of the 23 emotion dimensions, we fit a Ridge regression ($\alpha = 1.0$) predicting the emotion score from the 6-dimensional Witness Sim state vector, using 5-fold cross-validated predictions. The residual $\mathbf{r}_t \in \mathbb{R}^{23}$ is the component of the emotion signal not explained by the current state:

$$\mathbf{r}_t = \mathbf{s}_t - \hat{\mathbf{s}}_t, \qquad \hat{\mathbf{s}}_t = \mathrm{Ridge}(\mathbf{y}_t), \quad (11)$$

where $\mathbf{y}_t \in [0,1]^6$ is the state vector at turn t. Both feature sets were standardized to zero mean and unit variance before use.

3.3.2. Arc Similarity Analysis

Mean arc correlation. For each of the 10 synthetic archetypes, we computed the mean emotion arc across all witnesses by interpolating each transcript to $B = 20$ uniform bins and averaging. We then computed the Pearson correlation between each synthetic style's mean arc and the mean arc of the 55 real depositions, separately for each of the 23 emotions. The reported mean $r = 0.148$ is averaged across all emotion-style pairs.

Per-witness arc similarity. For each of the 10 sampled witnesses, we computed the real deposition arc $\tilde{\mathbf{S}}^{\text{real}} \in \mathbb{R}^{20\times 23}$ by averaging across all available real depositions for that witness, and the corresponding synthetic arc $\tilde{\mathbf{S}}^{\text{syn}} \in \mathbb{R}^{20\times 23}$ for each of the 10 archetypes. Each cell of the resulting (23×10) similarity matrix contains:

$$r_{e,a} = \text{PearsonR}\big(\tilde{s}^{\text{real}}_{:,e},\ \tilde{s}^{\text{syn}}_{:,e,a}\big)\,, \quad e \in \{1,\ldots,23\},\ a \in \{1,\ldots,10\}.$$

The best-fitting archetype is $a^* = \arg\max_a \frac{1}{23}\sum_e r_{e,a}$, and emotions are sorted by cross-archetype variance to highlight the most discriminating dimensions.

Permutation test. To assess whether the observed arc correlation reflects genuine temporal structure, we generated a null distribution by randomly permuting the 20 time bins of each synthetic mean arc while leaving the real arcs unchanged, repeating 1,000 times. This preserves marginal emotion-score distributions while destroying temporal ordering. We report the empirical p-value as the fraction of null correlations exceeding the observed value.

3.3.3. Real v Synthetic Deposition Analysis

PCA embedding. Each transcript (real and synthetic) was represented as a 460-dimensional arc vector $\mathbf{a} = \text{vec}(\tilde{\mathbf{S}}) \in \mathbb{R}^{460}$, standardized, and projected to two dimensions via PCA. We report the fraction of variance explained by PC1 and PC2, and visualize the separation between real and synthetic transcripts.

3.3.4. Timescale Analysis

Autocorrelation and decay fitting. For each of the 6 state dimensions and 23 emotion scores, we computed the mean autocorrelation function (ACF) at lags $k = 1,\ldots,20$ turns by averaging across all 100 synthetic transcripts. We then fit an exponential decay model to each mean ACF:

$$\text{ACF}(k) = e^{-k/\tau}, \tag{12}$$

using nonlinear least squares, yielding a characteristic decorrelation time τ (in turns) for each dimension. We report the range of τ values for state dimensions and emotion scores separately.

Mann-Whitney U test. To assess whether state and emotion dimensions operate on statistically distinct timescales, we applied a one-sided Mann-Whitney U test comparing the 6 state τ values to the 23 emotion τ values, testing the hypothesis that state dimensions have systematically larger τ.

Frequency decomposition. For each deposition, we decomposed each emotion time series into a slow component (rolling mean, window $= 5$ turns) and a fast component (residual from rolling mean). We then computed mean absolute Pearson correlations across all turn-level data, producing a 2×2 table of (state $\times$ {slow, fast}) and (residual $\times$ {slow, fast}) correlations.

3.3.5. Event Detection

Events and base rates. Binary event labels were extracted from the per-turn delta logs. Witness Combative marks turns where $A_t < 0.28 \wedge R_t > 0.60$ (base rate 21.4%, 14,628 positive turns). Personality Shift marks turns where $\|\mathbf{s}_t - \mathbf{s}_0\|_\infty > 0.35$ (base rate 0.2%, 112 positive turns).

Per-witness event-aligned averaging. Following the above, for each synthetic witness transcript, we identified all turns at which a Witness Combative event fired — defined as turns where the simulated state satisfies $A_t < 0.28 \wedge R_t > 0.60$ simultaneously. We then extracted a window of ± 5 turns centered on each event and averaged the raw emotion scores across all such windows for that witness. To aid interpretability, we restricted visualization to six emotion dimensions: *calm*, *defiant*, *hostile*, *angry*, *proud*, and *anxious*. This procedure was repeated independently for each witness, yielding one averaged emotion trajectory per witness around the event onset.

Classification. For each event and each feature set (emotion, residual), we trained a logistic regression classifier (5-fold stratified CV, $C = 1.0$, balanced class weights) to predict event occurrence from the 23 features at that turn. Out-of-fold predicted probabilities were collected across all folds and evaluated with AUC.

DeLong test. We compared the emotion AUC and residual AUC for each event using DeLong's test for paired AUC comparison (DeLong et al., 1988), which computes the variance of the AUC difference analytically via structural components of the ROC curve. We report the z-statistic and two-sided p-value.

4. Observations

4.1. Arc Similarity Analysis

Figure 1 shows representative per-witness arc similarity heatmaps, where columns are archetypes sorted best-to-worst by mean r across all 23 emotions, and rows are emotions sorted by cross-archetype variance. Best-fit archetypes vary across witnesses. (See Appendix for additional per witness heatmaps). In each case, the best-fit archetype shows a consistent advantage across the most discriminating emotions, while lower-ranked archetypes progressively lose signal or show negative correlations.

Mean correlation alone does not fully characterize similarity quality. The heatmaps additionally reveal consistency across multiple emotion dimensions, with the strongest-fitting archetypes exhibiting coherent positive correlations across the highest-variance emotions.

A permutation test confirms this temporal structure: shuffling the 20 time bins of each synthetic arc while preserving marginal emotion distributions produces a null distribution centered at zero, and the observed mean $r = 0.148$ exceeds all 1,000 permutations ($p < 0.001$).

Although the absolute effect size is modest, the consistency of the signal across emotions and witnesses, combined with the permutation results, indicates that the similarity is unlikely to arise from chance temporal alignment alone.

Figure 2 shows the real and best-fit synthetic arc for Jeffrey Kilper side by side. The synthetic trajectory reproduces several coarse temporal patterns observed in the real deposition, including aligned rises and declines across multiple emotion dimensions.

However, emotion magnitudes are systematically compressed in the synthetic transcript — real scores range from 0.05 to 0.30, while synthetic scores span roughly 0.00 to 0.08.

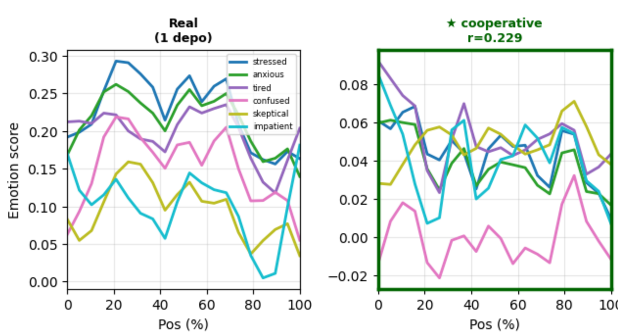

Figure 2. Real vs. best-fit synthetic arc for Jeffrey Kilper (cooperative, $r = 0.229$).

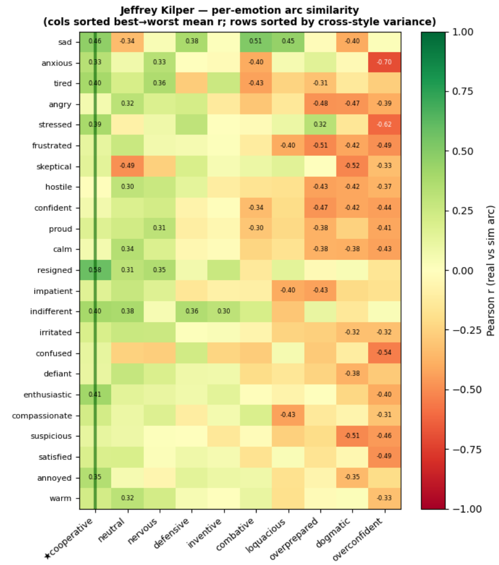

Figure 1. Per-witness arc similarity heatmap for Jeffrey Kilper. ($\star$ = best fit); Green indicates positive Pearson r between real and synthetic arc; red indicates negative.

4.2. Synthetic v. Real Deposition Similarity

PCA of the flattened 460-dimensional arc vectors (Figure 3) shows that real and synthetic transcripts occupy largely non-overlapping regions of the first principal component (PC1 = 29.9% variance). Synthetic transcripts cluster tightly on the negative PC1 side while real transcripts spread across positive PC1 values.

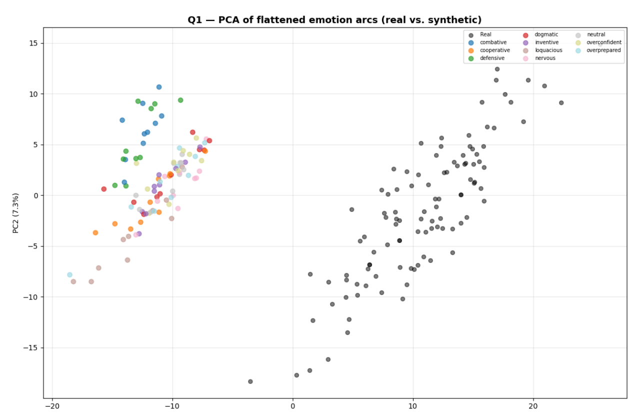

Figure 3. PCA of flattened 460-dimensional emotion arc vectors. Real transcripts (gray) and synthetic transcripts (colored by archetype).

4.3. Timescale Analysis

Exponential decay fitting to per-dimension autocorrelation functions yields characteristic decorrelation times τ of 17.7–525.1 turns for state dimensions and 0.8–2.6 turns for emotion scores.

A Mann-Whitney U test confirms no overlap between the two distributions ($U = 138$, $p < 0.001$). State dimensions encode slow, persistent behavioral dispositions; emotion scores fluctuate rapidly from turn to turn.

4.4. Event Detection

Figure 4 shows the event-aligned emotion trajectories around PERSONALITY SHIFT events ($n = 112$). A sharp spike in proud, confident, and enthusiastic scores occurs precisely at turn 0, followed by rapid decay, indicating that personality shifts are accompanied by a brief burst of assertive affect before the witness settles into a new behavioral regime. In contrast, the WITNESS COMBATIVE trajectories remain comparatively flat throughout the event window, both in aggregate and at the individual-deposition level.

Figure 4. PERSONALITY SHIFT event-aligned emotion trajectories ($n = 112$ events, ± 5 turns). Left: raw scores. Right: Δ vs. non-event baseline. Proud, confident, and enthusiastic spike sharply at turn 0.

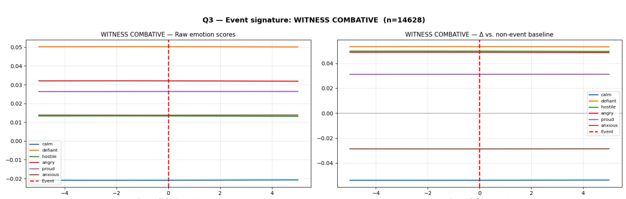

Figure 5. WITNESS COMBATIVE event-aligned emotion trajectories ($n = 14{,}628$ events). Flat lines across the window indicate that combative behavior reflects a stable personality disposition rather than a momentary escalation.

The DeLong test on paired AUC values (Table 2) reveals a clear asymmetry between the two feature sets. For WITNESS COMBATIVE, raw emotion features achieve AUC $= 0.996$ while residual features achieve only 0.707 ($\Delta = +0.289$, $z = 123.56$, $p < 0.001$): removing the slow state-aligned component substantially reduces predictive performance for sustained combative behavior.

For PERSONALITY SHIFT, the pattern reverses — residual features achieve AUC $= 0.963$ vs. 0.948 for raw emotion features ($\Delta = -0.015$, $z = -4.19$, $p < 0.001$). In contrast, residualization slightly improves detection of transient personality-shift events.

Event	Emo	Res	Δ	p
W. Combative	0.996	0.707	$+0.289$	< 0.001
Pers. Shift	0.948	0.963	-0.015	< 0.001

Table 2. DeLong paired AUC comparison (emotion vs. residual features). The sign of Δ flips between events, indicating complementary timescales.

Discussion

This work shows that deposition interactions contain recoverable temporal-affective structure and that this structure can be used to evaluate simulation realism.

Synthetic depositions reproduce real emotional structure. Synthetic and real depositions exhibit statistically aligned temporal emotion trajectories, with significance under permutation testing and meaningful variation in best-fit archetypes across witnesses. This suggests that Witness Sim captures aspects of real deposition dynamics. However, PCA still separates real from synthetic transcripts, and synthetic emotion magnitudes are compressed: real scores span roughly three times the range of synthetic scores. This gap reflects the limits of a low-dimensional state abstraction in modeling the emotional intensity of legal questioning.

Combativeness behaves like a mode, not an event. When Witness Sim labels a turn as WITNESS COMBATIVE, the witness's emotion arc shows little change before versus after the event. Furthermore, witnesses that are not the combative or defensive archetype never react or emotionally evolve in a way that triggers a WITNESS COMBATIVE event. Taken together, these observations suggest that combativeness is not usually a reactive turn-level disruption, but is instead a persistent state or characteristic.

These observations suggest that not all behavioral dynamics in depositions operate on the same timescale. While combativeness appears to reflect a persistent disposition, other behaviors may emerge as transient responses to specific lines of questioning. We investigate this possibility through the PERSONALITY SHIFT event.

Reactive affect is recoverable from residual emotion space. Beyond stable witness traits, depositions contain moments where questioning produces sharp affective shifts. The PERSONALITY SHIFT event captures one such transition: a spike in proud, confident, and enthusiastic emotion at turn 0, followed by rapid decay back to baseline rather than gradual drift. Residual emotion features, defined as the affective component not explained by the witness's chronic state, detect these moments better than raw emotion features alone (AUC $= 0.963$ vs. 0.948, $p < 0.001$). This suggests that emotion-vector analysis can identify not only who a witness is, but when something meaningful is happening in the exchange, with direct value for training attorneys to recognize and respond to affective vulnerability.

Together, these findings suggest that witness behavior is composed of both stable dispositions and transient reactions. Combativeness appears to function as a persistent behavioral mode, while personality shifts capture reactionary, short-lived departures from a witness's baseline state. Distinguishing between these timescales may be important for both simulation design and evaluation, as realistic simulated personas must capture both long-term behavioral consistency and short-term response to interpersonal interaction and external stimuli.

Limitations and Future Work

Several limitations suggest clear directions for future work. The real-deposition corpus comes from a single multidistrict litigation and includes only 55 transcripts across 10 witnesses, limiting both generalizability and statistical power. Because depositions were compressed into 20 temporal bins, future work could employ finer-grained temporal modeling to recover dynamics lost through aggregation.

Our emotion-vector construction also relies on a design choice inherited from prior work. Following Sofroniew et al., we extracted representations from layer 21, which was identified as a useful locus for affective structure in their analysis. However, that work was conducted on a different model family, whereas our analysis uses Llama-3.1-8B-Instruct due to its open-weight availability. Although layer 21 produced emotion vectors that qualitatively aligned with the target concepts and yielded meaningful downstream results, the optimal layer for emotion-vector extraction may differ across architectures. Future work should systematically evaluate layer choice across the model stack and compare alternative representation-extraction strategies to better characterize where affective and interpersonal concepts are most reliably encoded.

Incorporating audio and video modalities would further capture nonverbal cues that text-only analysis necessarily misses. Finally, human-subject evaluation with practicing attorneys could validate both realism and pedagogical value, calibrate the emotion-vector metric against expert judgments, and support fitting ODE coefficients and archetype attractors directly to real deposition trajectories. Learned pressure and sensitivity encoders could also replace keyword-based rules, improving robustness to paraphrase, indirect questioning, and domain transfer.

Conclusion

We introduced an emotion-vector evaluation framework for measuring behavioral realism in LLM-based legal simulation and applied it to Witness Sim across a corpus of real and synthetic deposition transcripts. Our results show that synthetic witnesses reproduce meaningful aspects of real deposition dynamics: emotion trajectories exhibit statistically significant temporal alignment with real testimony and vary systematically across witness archetypes. At the same time, synthetic depositions remain distinguishable from real ones and exhibit substantially compressed emotional intensity, highlighting important limitations in current simulation fidelity.

Beyond evaluating Witness Sim itself, this work reframes deposition realism as a dynamic behavioral problem rather than a static question-answering problem. For deposition training, a useful witness simulator must do more than accurately recall case facts: it must reproduce the behavioral consistency, emotional variability, and reactive dynamics that characterize real testimony under pressure. Our findings suggest that behavioral realism is both measurable and distinct from traditional notions of correctness, and that emotion-trajectory analysis provides one mechanism for quantifying this dimension of simulation fidelity and identifying where simulations succeed or fail.

More broadly, we argue that behavioral dynamics constitute an important and underexplored evaluation axis for LLM-based simulations. Realistic simulation depends not only on generating plausible responses, but on capturing how behavior unfolds through interaction with others. As language-model agents are increasingly deployed in training, education, negotiation, and other high-stakes interpersonal settings, evaluating how agents behave over the course of an interaction may become as important as evaluating the content of any individual response.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. As with any generative system, models trained on legal data carry inherent risks, including the potential to produce inaccurate, biased, or hallucinated outputs that may not reliably reflect statutes, case law, or jurisdictional nuance. In the legal domain, where errors can affect due process, client outcomes, and access to justice, such limitations warrant particular caution, and we emphasize that the methods presented here are intended to support, not substitute for, the judgment of qualified legal professionals.

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Appendix

A. Behavioral Justifications for State Dynamics

The functional form of each state update equation in Section 3.1.3 encodes a specific behavioral assumption. We list these here for completeness.

Table 3. Behavioral motivation for each state update.

State	Behavioral assumption
Composure (C)	Follows drain-and-recover dynamics, consistent with emotion-regulation depletion rather than a hard threshold model.
Knowledge (K)	Degrades under topic sensitivity and recovers only under low pressure, reflecting cognitive-load effects on working memory.
Agreeableness (A)	Moves away from its midpoint under pressure, allowing witnesses to polarize toward accommodation or hostility.
Verbosity (V)	Increases with topic sensitivity through over-explanation, but decreases under leading yes/no constraints.
Rigidity (R)	Accumulates under pressure and relaxes gradually once pressure decreases.
Performance (P)	Decays multiplicatively with fatigue ϕ_t, capturing faster degradation than recovery in over-prepared witnesses.

B. Per-Witness Arc Similarity: Full Results

B.1. Archetype Heatmaps

Figure 1 in the main paper shows the heatmap for Jeffrey Kilper. The remaining nine witnesses are shown below. Each heatmap displays Pearson r between the real deposition arc and each synthetic archetype arc, per emotion. Columns are sorted best-to-worst by mean r across all 23 emotions ($\star$ = best fit); rows are sorted by cross-archetype variance.

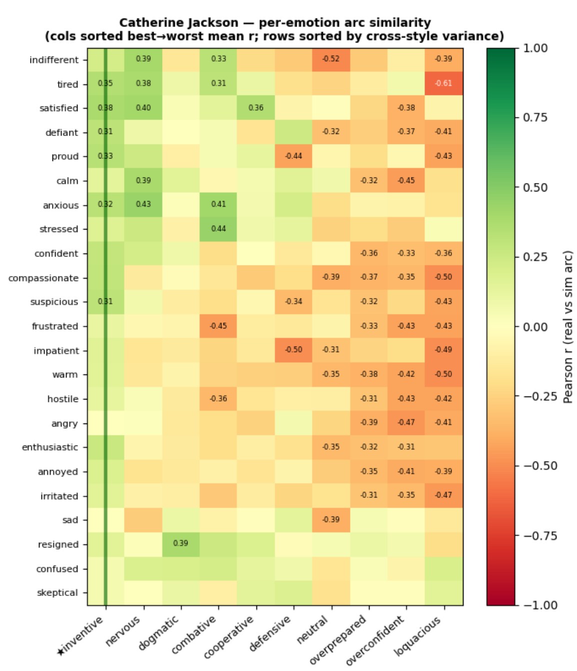

Figure 6. Catherine Jackson — best fit: *inventive*

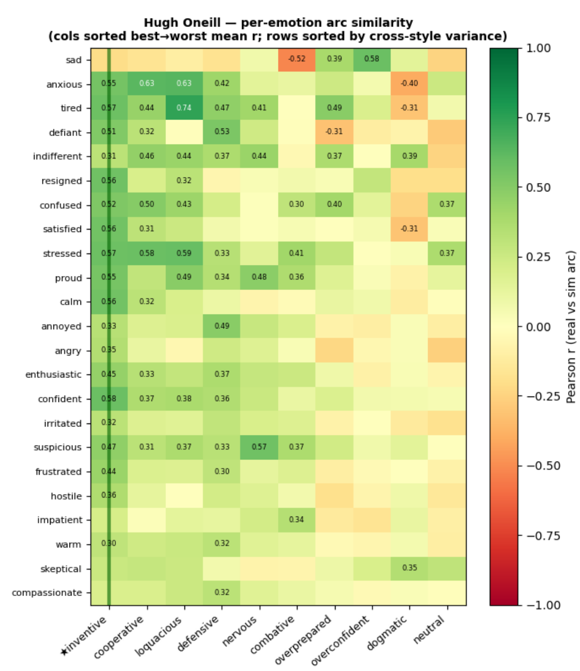

Figure 7. Hugh O'Neill — best fit: *inventive*

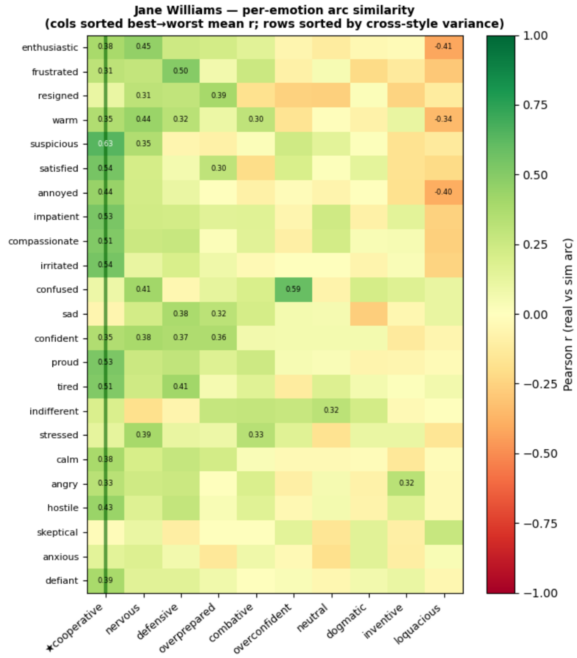

Figure 8. Jane Williams — best fit: *cooperative*

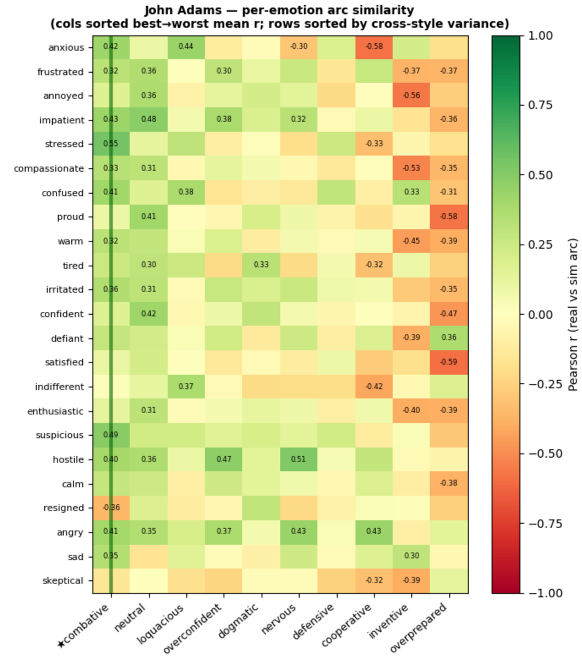

Figure 9. John Adams — best fit: *combative*

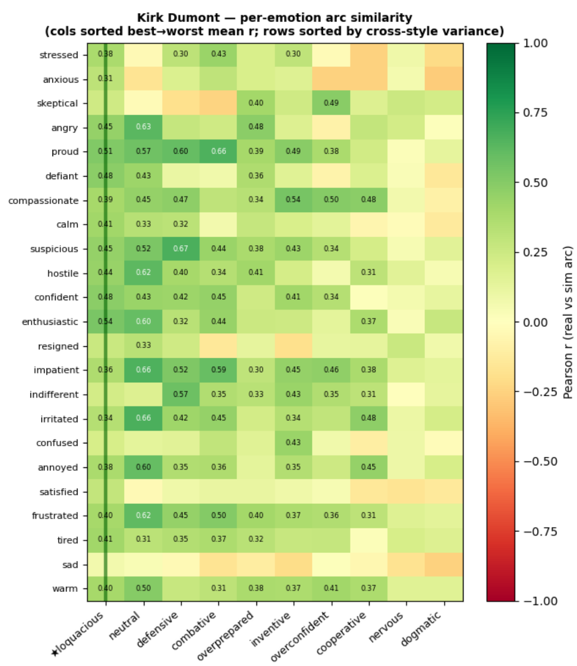

Figure 10. Kirk Dumont — best fit: *loquacious*

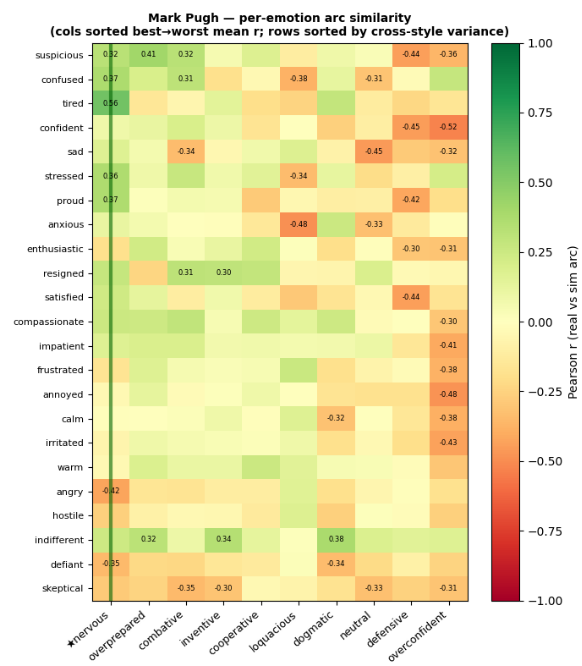

Figure 11. Mark Pugh — best fit: *nervous*

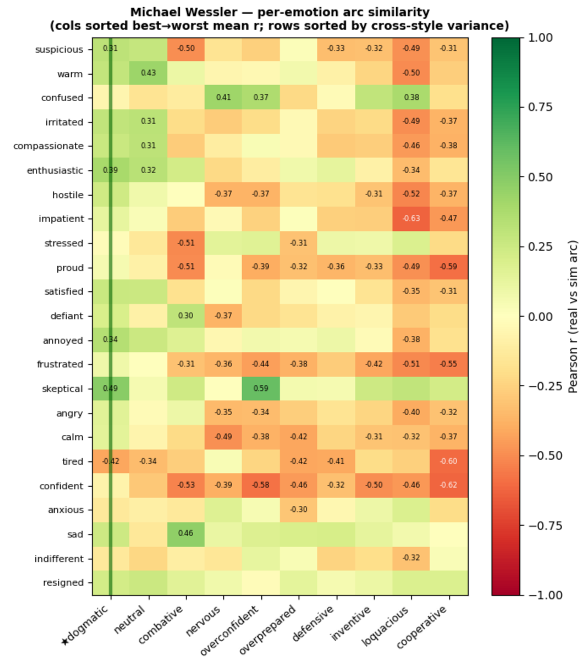

Figure 12. Michael Wessler — best fit: *dogmatic*

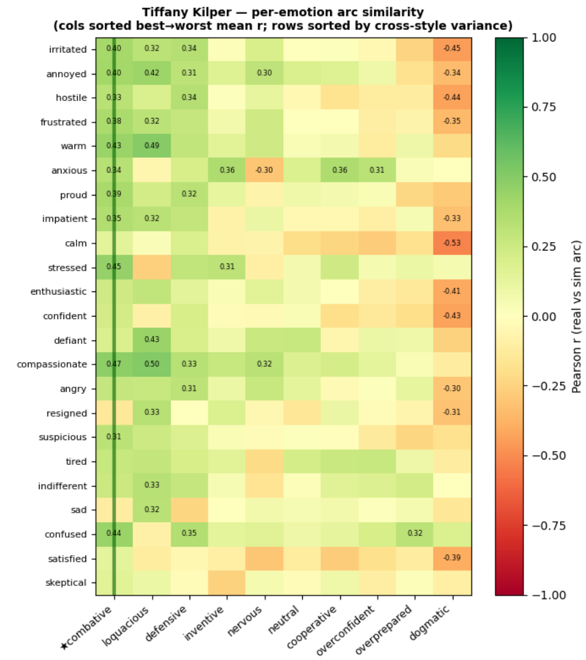

Figure 13. Tiffany Kilper — best fit: *combative*

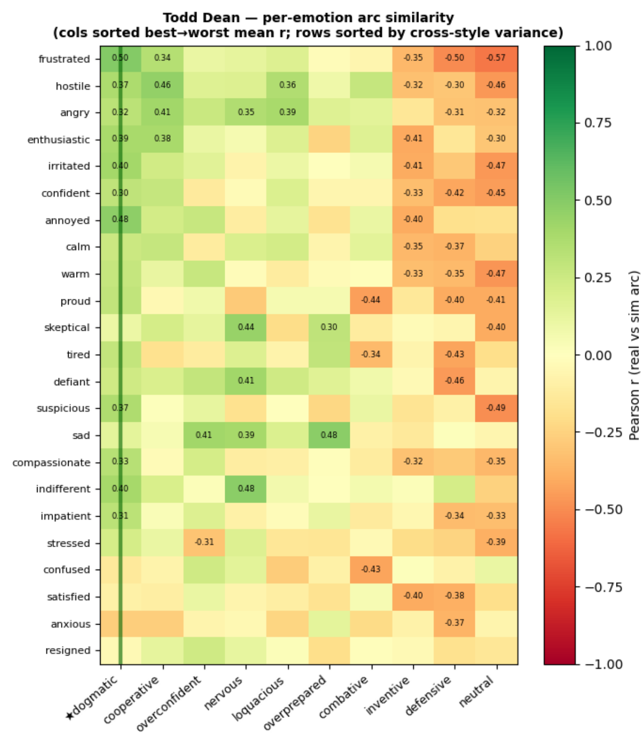

Figure 14. Todd Dean — best fit: *dogmatic*

B.2. Emotion Arc Trajectories

Figure 2 in the main paper shows the emotion trajectory of the real deposition vs the best fit for Jeffery Kilper. The remaining nine witnesses

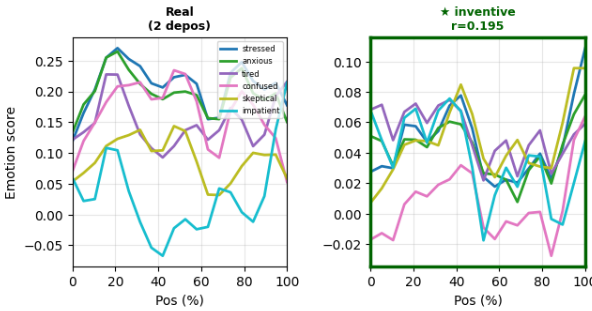

Figure 15. Catherine Jackson — best fit: *inventive*

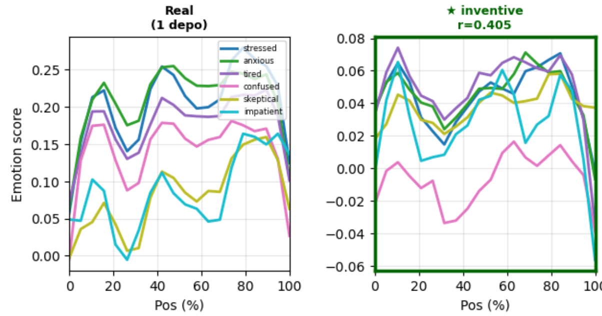

Figure 16. Hugh O'Neill — best fit: *inventive*

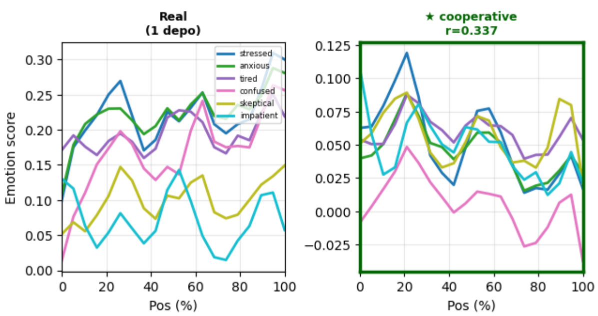

Figure 17. Jane Williams — best fit: *cooperative*

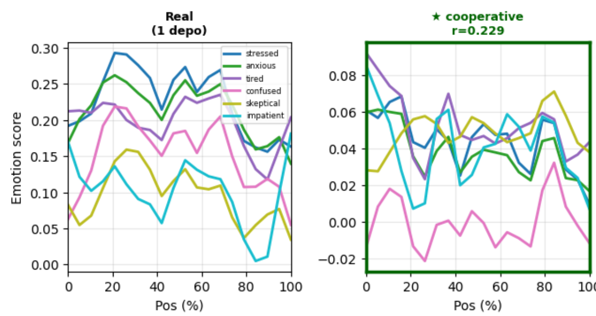

Figure 18. John Adams — best fit: *combative*

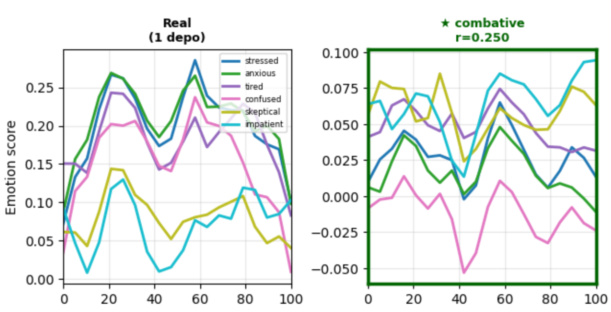

Figure 19. Kirk Dumont — best fit: *loquacious*

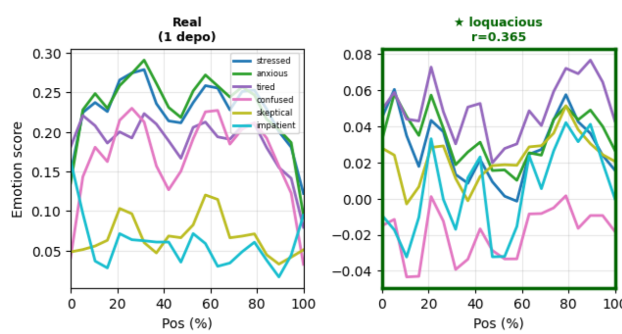

Figure 20. Mark Pugh — best fit: *nervous*

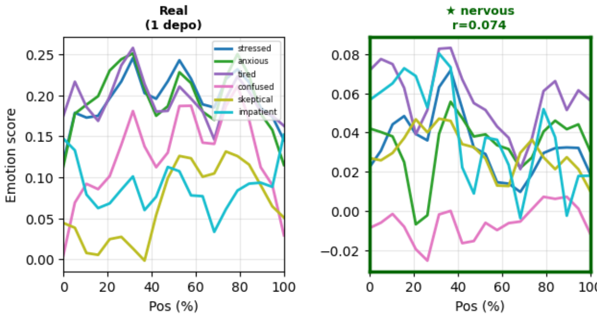

Figure 21. Michael Wessler — best fit: *dogmatic*

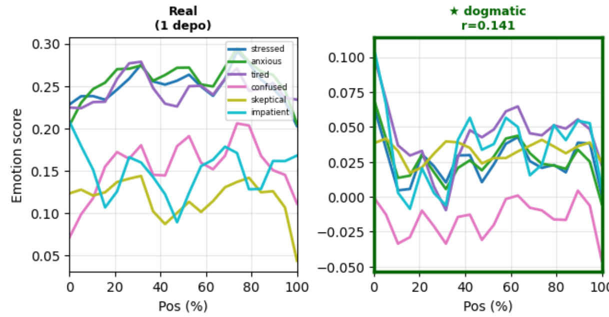

Figure 22. Tiffany Kilper — best fit: *combative*

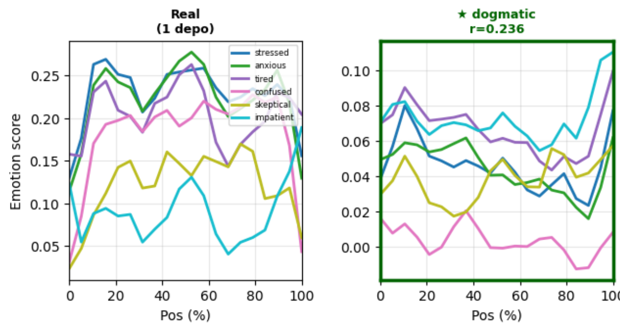

Figure 23. Todd Dean — best fit: *dogmatic*

C. Permutation Test: Arc Correlation Above Chance

Figure 24 shows the null distribution of mean arc correlations generated by shuffling the 20 time bins of each synthetic arc 1,000 times while leaving real arcs unchanged. The observed mean $r = 0.1477$ (red line) far exceeds both the null mean of 0.0002 and the null 95th percentile of 0.0605, confirming that the temporal structure shared between real and synthetic arcs is not attributable to chance alignment of marginal emotion distributions.

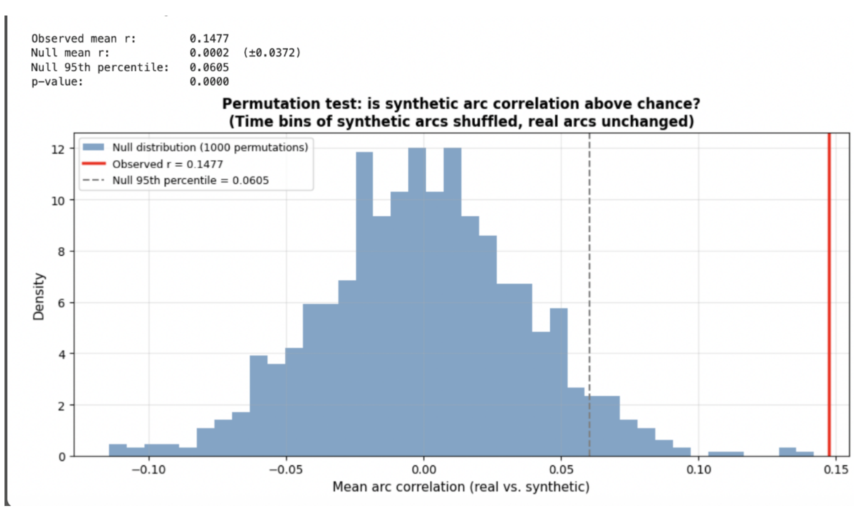

Figure 24. Permutation test for arc correlation. Blue histogram shows the null distribution (1,000 permutations of synthetic arc time bins). Red line: observed $r = 0.1477$. Dashed line: null 95th percentile $= 0.0605$. Empirical $p < 0.001$.

D. Residual Emotion Space Structure

Figure 25 shows the PCA loadings for the top four principal components of the residual emotion space — the component of emotion scores not explained by the 6D state vector.

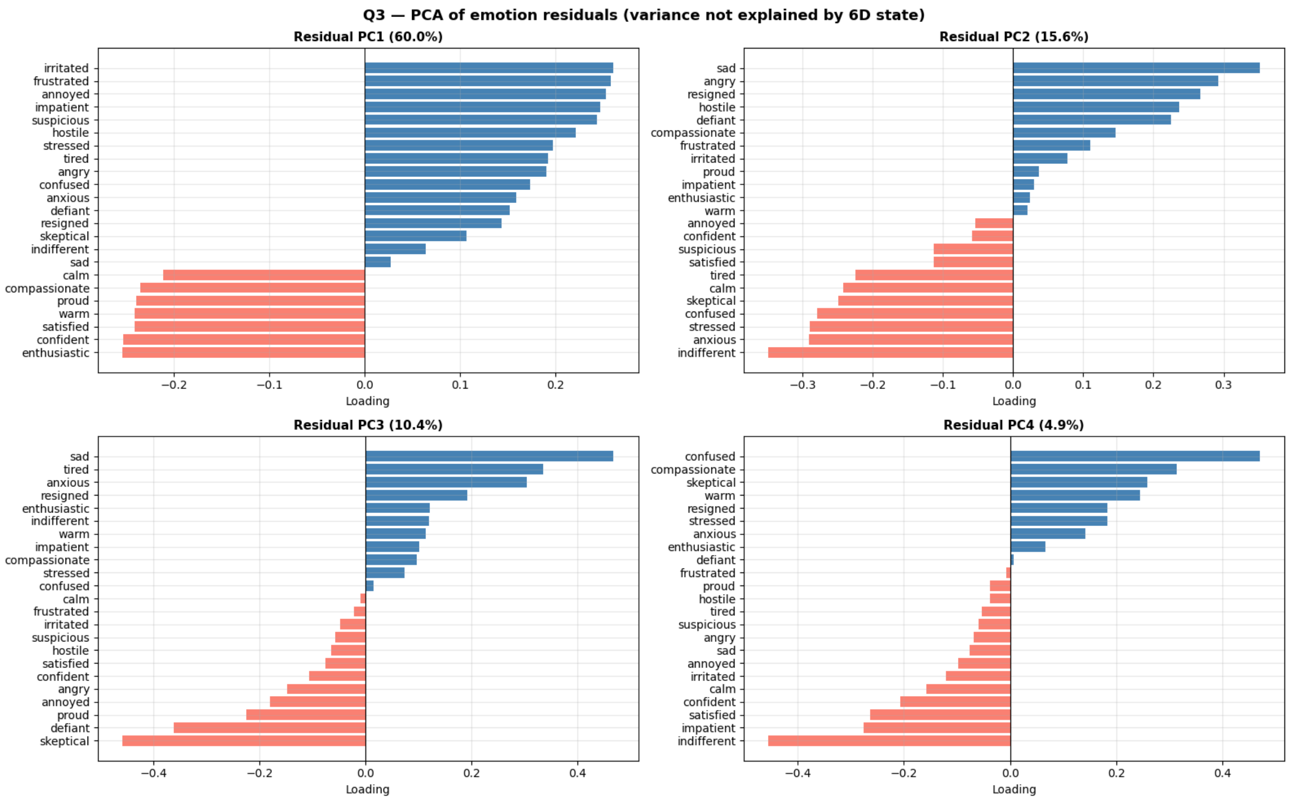

Figure 25. PCA loadings for the top four components of the residual emotion space. PC1 (46.8%) and PC2 (15.8%) capture the largest independent affective dimensions not explained by the state vector.

D.1.

E. Emotion Vector Construction: Prompts and Data

The emotion vector construction methodology follows Sofroniew et al. (2026), who demonstrate that directional emotion vectors can be extracted from transformer residual streams by contrasting mean activations across emotionally evocative stories. We replicate their story generation procedure with the following emotions, topics, and prompt template.

E.1. Emotion List

The following 23 emotions were used to construct directional emotion vectors. Each emotion was evoked through 125 generated stories across 25 topically diverse scenarios.

anxious	hostile	irritated	satisfied	warm
confident	angry	annoyed	calm	confused
indifferent	compassionate	defiant	sad	resigned
enthusiastic	tired	frustrated	impatient	stressed
suspicious	proud	skeptical		

E.2. Story Generation Topics

Stories were generated across the following 25 topically diverse scenarios to prevent topic-emotion confounding. Each (emotion, topic) pair produced 5 stories, yielding 2,875 stories total.

1. A person discovers their old love letters have been turned into a stage play
2. A coworker quietly takes credit for a group project during a company meeting
3. A parent finds out their child has been skipping therapy sessions
4. Someone learns their identical twin has been impersonating them online
5. A person receives a voicemail meant for someone who died
6. A gardener finds a neighbor has been harvesting their vegetable garden
7. A person discovers their estranged sibling lives two streets away
8. Someone finds out their closest friend testified against them without telling them
9. A person's dog recognizes a stranger in a way that suggests a prior connection
10. A teenager finds their parent's old arrest record
11. A person receives a corrected version of their own memory from a therapist's notes
12. Someone learns their wedding venue has been double-booked
13. A person discovers their child has been secretly supporting a distant relative
14. An employee finds out their resignation letter was never submitted by their manager
15. A person learns the eulogy they wrote for a funeral was significantly rewritten
16. Someone finds a photograph of themselves at a place they have no memory of visiting
17. A person's handwritten recipe is printed on a mass-produced product without credit
18. A coworker confesses they have been covering for a mutual colleague's absences
19. Someone discovers their landlord has been entering the apartment unannounced
20. A parent learns their adult child has quietly paid off a family debt
21. A person realizes their therapist and their boss know each other socially
22. Someone finds their name listed as a dedication in a stranger's published memoir
23. A person learns their childhood nickname became a running joke among relatives
24. Two old friends realize they were in the same hospital on the same night years ago
25. A person discovers their long-term pen pal has been using a fictitious name

E.3. Story Generation Prompt

The following prompt structure follows Sofroniew et al. (2026). Stories were generated via `claude-opus-4.5` for each (emotion, topic) pair. The key design constraint is that the target emotion is never named directly — it must be conveyed through behavior, physiology, dialogue, and context only.

```
Write {n_stories} different stories based on the following premise.

Topic: {topic}

The story should follow a character who is feeling {emotion}.

Format the stories like so:

[story 1]
[story 2]
[story 3]

etc.

The paragraphs should each be a fresh start, with no continuity. Try to make them
diverse and not use the same turns of phrase. Across the different stories, use a
mix of third-person narration and first-person narration.

IMPORTANT: You must NEVER use the word '{emotion}' or any direct synonyms of it in
the stories. Instead, convey the emotion ONLY through:

 • The character's actions and behaviors
 • Physical sensations and body language
 • Dialogue and tone of voice
 • Thoughts and internal reactions
 • Situational context and environmental descriptions

The emotion should be clearly conveyed to the reader through these indirect means,
but never explicitly named.
```